

SEMICONDUCTOR MISSION AND ELECTRONICS MANUFACTURING

DESIGN-TO-DEVICE RAIL → SEMICONDUCTOR BUSINESS-MODEL MATRIX → FRONT-END VERSUS BACK-END → NODE AND LITHOGRAPHY LADDER → ADVANCED INTEGRATION MAP → MATERIALS TAXONOMY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 DESIGN-TO-DEVICE RAIL

DESIGN-TO-DEVICE RAIL ARCHITECTURE, VERIFICATION AND TAPE-OUT TRANSISTOR AND INTERCONNECT LAYERS ON WAFER WAFER BECOMES INDIVIDUAL DIES ASSEMBLE, TEST, MARK AND PACKAGE DOWNSTREAM ELECTRONICS INTEGRATION

DESIGN / IP / EDA → architecture, verification and tape-out → FOUNDRY / FAB → transistor and interconnect layers on wafer → DICING → wafer becomes individual dies → ATMP / OSAT → assemble, test, mark and package → BOARD / MODULE / DEVICE → downstream electronics integration

STARTING CONDITION / INPUT

- DESIGN / IP / EDA → architecture, verification and tape-out
- FOUNDRY / FAB → transistor and interconnect layers on wafer

SCIENTIFIC OR ENGINEERING MECHANISM

- DICING → wafer becomes individual dies
- ATMP / OSAT → assemble, test, mark and package

OUTPUT / FAILURE BOUNDARY

- BOARD / MODULE / DEVICE → downstream electronics integration
- DESIGN / IP / EDA → architecture, verification and tape-out

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BOARD / MODULE / DEVICE → downstream electronics integration.

01

STAGE 01 SEMICONDUCTOR BUSINESS-MODEL MATRIX

SEMICONDUCTOR BUSINESS-MODEL MATRIX PHYSICAL WAFER-MANUFACTURING PLANT DESIGNS AND OUTSOURCES WAFER MANUFACTURE MANUFACTURES TO CUSTOMERS' DESIGNS DESIGNS AND FABRICATES ITS OWN PRODUCTS PLANT TYPE AND FIRM MODEL ANSWER DIFFERENT QUESTIONS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
FAB → physical wafer-manufacturing plant	FOUNDRY → manufactures to customers' designs	RULE → plant type and firm model answer different questions
FABLESS → designs and outsources wafer manufacture	IDM → designs and fabricates its own products	FAB → physical wafer-manufacturing plant

SYSTEM / DEVICE / INSTITUTION

- FAB → physical wafer-manufacturing plant
- FABLESS → designs and outsources wafer manufacture

DECISIVE TECHNICAL DISTINCTION

- FOUNDRY → manufactures to customers' designs
- IDM → designs and fabricates its own products

STATUS / UNIT / EVIDENCE LIMIT

- RULE → plant type and firm model answer different questions
- FAB → physical wafer-manufacturing plant

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → plant type and firm model answer different questions.

02

STAGE 02 FRONT-END VERSUS BACK-END

FRONT-END VERSUS BACK-END FRONT END DEPOSITION, LITHOGRAPHY, ETCHING, DOPING, METALLISATION PROCESSED WAFER CONTAINING DIES BACK END BONDING, ENCAPSULATION, TESTING, MARKING, PACKAGING PROCESS DESCRIPTION; OSAT OUTSOURCED SERVICE MODEL

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
FRONT END → deposition, lithography, etching, doping, metallisation	BACK END → bonding, encapsulation, testing, marking, packaging	TRAP → back-end capacity is not front-end fabrication
OUTPUT → processed wafer containing dies	ATMP → process description; OSAT → outsourced service model	FRONT END → deposition, lithography, etching, doping, metallisation

SYSTEM / DEVICE / INSTITUTION

- FRONT END → deposition, lithography, etching, doping, metallisation
- OUTPUT → processed wafer containing dies

DECISIVE TECHNICAL DISTINCTION

- BACK END → bonding, encapsulation, testing, marking, packaging
- ATMP → process description; OSAT → outsourced service model

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → back-end capacity is not front-end fabrication
- FRONT END → deposition, lithography, etching, doping, metallisation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → back-end capacity is not front-end fabrication.

03

STAGE 03 NODE AND LITHOGRAPHY LADDER

NODE AND LITHOGRAPHY LADDER BROAD AUTOMOTIVE, INDUSTRIAL AND POWER USES MATURE AND MID-RANGE LITHOGRAPHY ROUTE LEADING EDGE RISING COMPLEXITY AND CAPITAL INTENSITY CONCENTRATED EQUIPMENT AND EXPORT-CONTROL CHOKEPOINT

VERIFY EVERY PROJECT-SPECIFIC NODE CLAIM OFFICIALLY

DUV → mature and mid-range lithography route → LEADING EDGE → rising complexity and capital intensity → EUV → concentrated equipment and export-control chokepoint

03

STAGE 03 NODE AND LITHOGRAPHY LADDER

NODE AND LITHOGRAPHY LADDER

BROAD AUTOMOTIVE, INDUSTRIAL AND POWER USES

MATURE AND MID-RANGE LITHOGRAPHY ROUTE

LEADING EDGE

RISING COMPLEXITY AND CAPITAL INTENSITY

CONCENTRATED EQUIPMENT AND EXPORT-CONTROL CHOKEPOINT

VERIFY EVERY PROJECT-SPECIFIC NODE CLAIM OFFICIALLY

DUV → mature and mid-range lithography route

LEADING EDGE → rising complexity and capital intensity

EUV → concentrated equipment and export-control chokepoint

STARTING CONDITION / INPUT

- MATURE / MID-RANGE → broad automotive, industrial and power uses
- DUV → mature and mid-range lithography route

SCIENTIFIC OR ENGINEERING MECHANISM

- LEADING EDGE → rising complexity and capital intensity
- EUV → concentrated equipment and export-control chokepoint

OUTPUT / FAILURE BOUNDARY

- RULE → verify every project-specific node claim officially
- MATURE / MID-RANGE → broad automotive, industrial and power uses

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → verify every project-specific node claim officially.

04

STAGE 04 ADVANCED INTEGRATION MAP

ADVANCED INTEGRATION MAP

SPECIALISED DIES COMBINED AS ONE SYSTEM

DENSER PACKAGE-LEVEL INTEGRATION

THROUGH-SILICON VIAS

VERTICAL INTERCONNECT ROUTE

HETEROGENEOUS INTEGRATION

COMBINE UNLIKE FUNCTIONS OR PROCESSES

PERFORMANCE GAINS NEED NOT BEGIN WITH SMALLER LITHOGRAPHY

2.5D / 3D → denser package-level integration

THROUGH-SILICON VIAS → vertical interconnect route

HETEROGENEOUS INTEGRATION → combine unlike functions or processes

CONCEPT / ARCHITECTURE

- CHIPLETS → specialised dies combined as one system
- 2.5D / 3D → denser package-level integration

MECHANISM / APPLICATION

- THROUGH-SILICON VIAS → vertical interconnect route
- HETEROGENEOUS INTEGRATION → combine unlike functions or processes

READINESS / STATUS LIMIT

- INSIGHT → performance gains need not begin with smaller lithography
- CHIPLETS → specialised dies combined as one system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INSIGHT → performance gains need not begin with smaller lithography.

05

STAGE 05 MATERIALS TAXONOMY

MATERIALS TAXONOMY

COMPOUND SEMICONDUCTORS

APPLICATION FAMILIES

SILICON PHOTONICS

SILICON OPTICAL-INTERCONNECT PLATFORM

POLICY GROUPING

MAY PLACE UNLIKE TECHNOLOGIES TOGETHER

SCHEME NOMENCLATURE DOES NOT CHANGE MATERIAL PHYSICS

POWER / RF / OPTOELECTRONICS → application families

SILICON PHOTONICS → silicon optical-interconnect platform

POLICY GROUPING → may place unlike technologies together

CONCEPT / ARCHITECTURE

- COMPOUND SEMICONDUCTORS → GaN / GaAs / InP / SiC
- POWER / RF / OPTOELECTRONICS → application families

MECHANISM / APPLICATION

- SILICON PHOTONICS → silicon optical-interconnect platform
- POLICY GROUPING → may place unlike technologies together

READINESS / STATUS LIMIT

- RULE → scheme nomenclature does not change material physics
- COMPOUND SEMICONDUCTORS → GaN / GaAs / InP / SiC

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → scheme nomenclature does not change material physics.

06

STAGE 06 FAB VIABILITY CHAIN

FAB VIABILITY CHAIN

RELIABLE POWER + ULTRA-PURE WATER

STABLE PROCESSING

GASES + CHEMICALS + EQUIPMENT

CONTROLLED PROCESS STEPS

REPEATED RUNS

YIELD LEARNING

RELIABILITY EVIDENCE

GASES + CHEMICALS + EQUIPMENT → controlled process steps

REPEATED RUNS → yield learning

RELIABILITY EVIDENCE → customer qualification

STARTING CONDITION / INPUT

- RELIABLE POWER + ULTRA-PURE WATER → stable processing

SCIENTIFIC OR ENGINEERING MECHANISM

- REPEATED RUNS → yield learning

OUTPUT / FAILURE BOUNDARY

- OUTCOME → inauguration alone is not competitive production

06

STAGE 06 FAB VIABILITY CHAIN

FAB VIABILITY CHAIN

RELIABLE POWER + ULTRA-PURE WATER

STABLE PROCESSING

GASES + CHEMICALS + EQUIPMENT

CONTROLLED PROCESS STEPS

REPEATED RUNS

YIELD LEARNING

RELIABILITY EVIDENCE

RELIABLE POWER + ULTRA-PURE WATER → stable processing

GASES + CHEMICALS + EQUIPMENT → controlled process steps

REPEATED RUNS → yield learning

RELIABILITY EVIDENCE → customer qualification

STARTING CONDITION / INPUT

- RELIABLE POWER + ULTRA-PURE WATER → stable processing
- GASES + CHEMICALS + EQUIPMENT → controlled process steps

SCIENTIFIC OR ENGINEERING MECHANISM

- REPEATED RUNS → yield learning
- RELIABILITY EVIDENCE → customer qualification

OUTPUT / FAILURE BOUNDARY

- OUTCOME → inauguration alone is not competitive production
- RELIABLE POWER + ULTRA-PURE WATER → stable processing

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → inauguration alone is not competitive production.

07

STAGE 07 INSTITUTION AND SCHEME MAP

INSTITUTION AND SCHEME MAP

PARENT MINISTRY

IMPLEMENTING AGENCY

SEMICON INDIA

SCHEME PACKAGE

DESIGN-FOCUSED CHANNEL

SEPARATE COMPLEMENTARY ELECTRONICS INCENTIVES

MeitY → parent ministry

ISM → implementing agency

SEMICON INDIA → scheme package

DLI → design-focused channel

CONCEPT / ARCHITECTURE

- MeitY → parent ministry
- ISM → implementing agency

MECHANISM / APPLICATION

- SEMICON INDIA → scheme package
- DLI → design-focused channel

READINESS / STATUS LIMIT

- SPECS / PLI → separate complementary electronics incentives
- MeitY → parent ministry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPECS / PLI → separate complementary electronics incentives.

08

STAGE 08 SEMICON 1.0 TO 2.0

SEMICON 1.0 TO 2.0

SEMICON INDIA 1.0

RS 76,000 CRORE UMBRELLA

FOUR CHANNELS

FABS, DISPLAYS, COMPOUND/SENSOR/ATMP, DLI

15 JUL 2026

SEMICON 2.0 CABINET APPROVAL

SEMICON 2.0

SEMICON INDIA 1.0 → Rs 76,000 crore umbrella

FOUR CHANNELS → fabs, displays, compound/sensor/ATMP, DLI

15 JUL 2026 → Semicon 2.0 Cabinet approval

SEMICON 2.0 → Rs 1,27,500 crore approved outlay

SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent

CONCEPT / ARCHITECTURE

- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella
- FOUR CHANNELS → fabs, displays, compound/sensor/ATMP, DLI

MECHANISM / APPLICATION

- 15 JUL 2026 → Semicon 2.0 Cabinet approval
- SEMICON 2.0 → Rs 1,27,500 crore approved outlay

READINESS / STATUS LIMIT

- SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent
- SEMICON INDIA 1.0 → Rs 76,000 crore umbrella

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIX PILLARS → design, tools/materials, fabs, back end, R&D, talent.

09

STAGE 09 FACILITY EVIDENCE MAP

FACILITY EVIDENCE MAP

FAB; LATER NAMED COMPOUND-FAB PLUS ATMP APPROVAL

OFFICIALLY IDENTIFIED OSAT

ATMP LOCATION

OFFICIALLY IDENTIFIED BACK-END CLUSTER AND INAUGURATIONS

SUCHI SEMICON OSAT APPROVAL

LOCATION MUST TRAVEL WITH PROJECT AND STATUS VERB

MUST REMEMBER

DHOLERA → fab; later named compound-fab plus ATMP approval

MORIGAON → officially identified OSAT / ATMP location

SANAND → officially identified back-end cluster and inaugurations

SURAT → Suchi Semicon OSAT approval

CONCEPT / ARCHITECTURE

- DHOLERA → fab; later named compound-fab plus ATMP approval
- MORIGAON → officially identified OSAT / ATMP location

MECHANISM / APPLICATION

- SANAND → officially identified back-end cluster and inaugurations
- SURAT → Suchi Semicon OSAT approval

READINESS / STATUS LIMIT

- RULE → location must travel with project and status verb
- MUST REMEMBER: The semiconductor chain separates design/IP, wafer fabrication, compound...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location must travel with project and status verb.

